

STAT 542 - Statistical Learning

Homework Three

Fall 2026, University of Illinois Urbana-Champaign

Due date: TBA

Total Points = 83 + 3 Bonus

1. Answer the following questions about Beta-Binomial inference.

- a) Suppose we are interested in the survival rate for a high-risk operation in a new hospital. Experience in other hospitals indicates that the probability of survival for a patient is expected to be around 0.9 and it would be fairly surprising if it were less than 0.8 and more than 0.97.

- i. If you were to use the Beta distribution to represent the opinion above, what values of α and β would you use, and why? **(3 points)**

Suppose now that $n = 10$ patients are operated on in this new hospital and all of them survived.

- i. What is the current posterior distribution of the survival rate? **(2 points)**
- ii. What is the probability that the next patient will survive the operation? **(3 points)**
- iii. What is the probability that there will be two or more deaths in the next 20 operations? **(4 points)**

- b) Suppose we have a coin that we know is strongly biased, either usually coming up heads or usually coming up tails, but we do not know which.

- i. Express this information in the form of a $\text{Beta}(\alpha, \beta)$ prior. What values of α and β would you use, and why? **(2 points)**
- ii. We now flip the coin 5 times and it comes up heads in 4 of the 5 flips. Based on your prior from the previous part, what is the posterior distribution? **(2 points)**

- c) Consider the Bayesian model

$$\theta \sim \text{Uniform}(0, 1), \quad X_i \mid \theta \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(\theta).$$

Prove that

$$\mathbb{P}\{X_{n+1} = 1 \mid X_1 = x_1, \dots, X_n = x_n\} = \frac{1 + \sum_{i=1}^n x_i}{n + 2}.$$

This is called the Laplace rule of succession. **(3 points)**

2. Consider the following Bayesian model:

$$\theta \sim \text{Gamma}(\alpha, \beta), \quad f(\theta) \propto \theta^{\alpha-1} e^{-\beta\theta} I\{\theta > 0\}, \quad Y | \theta \sim \text{Poisson}(k\theta).$$

Treat α, β, k as fixed positive constants.

- a) Prove that $\theta | Y = y \sim \text{Gamma}(\alpha + y, \beta + k)$. **(2 points)**
- b) Prove that the marginal distribution of Y is given by

$$\mathbb{P}\{Y = y\} = \frac{\Gamma(\alpha + y)}{\Gamma(\alpha)y!} \left(\frac{k}{\beta + k}\right)^y \left(\frac{\beta}{\beta + k}\right)^\alpha, \quad y = 0, 1, 2, \dots$$

(3 points)

3. Consider the following baseball dataset (from Efron) used in class. The goal is to use the players' early season performance (second column) to predict their end-of-season performance (third column). Use the model

$$\log a, \log b \stackrel{\text{i.i.d.}}{\sim} \text{Uniform}(-\infty, \infty), \quad \theta_i | a, b \stackrel{\text{i.i.d.}}{\sim} \text{Beta}(a, b), \quad H_i | \theta_i \stackrel{\text{ind}}{\sim} \text{Binomial}(45, \theta_i).$$

Here H_i is the number of hits in the first 45 at-bats for player i .

- a) Using a numerical discretization approach, implement Bayesian inference in this model and obtain posterior samples of $\theta_1, \dots, \theta_N$. Compute posterior mean estimates for each θ_i . Compute the accuracy of these estimates in the two metrics: sum of squares and sum of absolute values. Compare with the naive estimates $H_i/45$ and the James–Stein estimates computed in class. **(5 points)**
- b) Using the posterior samples, compute 95% credible intervals for each θ_i . Compare with the actual values of θ_i . How many of the actual values lie in these intervals? **(3 points)**

Player Name	Hits in first 45 at-bats	End-of-season average
Clemente	18	.346
F Robinson	17	.298
F Howard	16	.276
Johnstone	15	.222
Berry	14	.273
Spencer	14	.270
Kessinger	13	.263
L Alvarado	12	.210
Santo	11	.269
Swoboda	11	.230
Unser	10	.264
Williams	10	.256
Scott	10	.303
Petrocelli	10	.264
E Rodriguez	10	.226
Campaneris	9	.286
Munson	8	.316
Alvis	7	.200

4. Suppose $(X_1, Y_1), \dots, (X_n, Y_n)$ are i.i.d. observations from a distribution with correlation ρ .

- a) Describe an algorithm based on classical bootstrap to obtain a 95% interval for ρ . **(3 points)**

- b) Describe an algorithm based on Bayesian bootstrap to obtain a 95% interval for ρ . **(3 points)**
 - c) Consider the data in `PearsonHeightData.txt`, which contains heights of adult father-son pairs. Using the methods above, compute 95% intervals for ρ . Compare the resulting intervals: are they similar? Which method produces the widest interval? **(6 points)**
5. Consider the text data file `alice.txt`. The goal is to replicate the analysis from class for this dataset using i.i.d. and AR(1) models.
- a) Tokenize `alice.txt` so that each token corresponds to a single word, with the period (full stop) treated as a separate token. **(1 point)**
 - b) Construct a vocabulary consisting of all unique tokens in the corpus. Let K denote the vocabulary size. **(1 point)**
 - c) The first model assumes $y_1, \dots, y_n$ i.i.d. from $(p_1, \dots, p_K)$, where y_i is the i th token. Use the prior

$$(p_1, \dots, p_K) \sim \text{Dirichlet}(\alpha, \dots, \alpha), \quad \alpha = 10^{-3}.$$

Calculate the posterior distribution of $(p_1, \dots, p_K)$.

- i. Output the five words with the largest posterior mean estimates $\hat{p}_k$. Comment on whether this is reasonable given the nature of the text. **(2 points)**
 - ii. Generate $s = 20$ sentences by sequentially sampling tokens until a period token is produced. Comment on how realistic the generated sentences appear. **(3 points)**
- d) The second model is an AR(1) model with likelihood

$$\prod_{i=1}^K \prod_{j=1}^K p_{j|i}^{x_{j|i}},$$

where $x_{j|i}$ counts how often token j follows token i , and $p_{j|i}$ is the probability of token j given previous token i . Use independent row priors

$$(p_{j|i} : j = 1, \dots, K) \stackrel{\text{i.i.d.}}{\sim} \text{Dirichlet}(a_1, \dots, a_K).$$

Using maximum marginal likelihood, estimate $a_1, \dots, a_K$. **(5 points)**

- e) Using the estimated $a_1, \dots, a_K$, calculate posterior mean estimates $\hat{p}_{j|i}$ for all i . For each of the tokens `alice`, `she`, and `the`, report the four tokens j with the largest $\hat{p}_{j|i}$. **(3 points)**
 - f) Use the fitted AR(1) model to generate $s = 20$ sentences, each starting with the period token and stopping at the next period token. Comment on realism and compare with the i.i.d. sentences. **(4 points)**
6. Consider two binary variables A and B . We observe $(A_1, B_1), \dots, (A_n, B_n)$ from n subjects, where each pair is one of $(0, 0), (0, 1), (1, 0), (1, 1)$ and the pairs are i.i.d. Write

$$\theta_{ab} = \mathbb{P}\{A_i = a, B_i = b\}, \quad a, b \in \{0, 1\}.$$

We consider two models for $(\theta_{00}, \theta_{01}, \theta_{10}, \theta_{11})$.

- a) Model 1 uses $\mathbb{P}\{A_i = a, B_i = b\} = \mathbb{P}\{A_i = a\}\mathbb{P}\{B_i = b \mid A_i = a\}$, with

$$\mathbb{P}\{A_i = 1\}, \mathbb{P}\{B_i = 1 \mid A_i = 1\}, \mathbb{P}\{B_i = 1 \mid A_i = 0\} \stackrel{\text{i.i.d.}}{\sim} \text{Uniform}(0, 1).$$

Write down the joint density on $(\theta_{00}, \theta_{01}, \theta_{10}, \theta_{11})$ under Model 1. (**3 points**)

- b) Model 2 uses $\mathbb{P}\{A_i = a, B_i = b\} = \mathbb{P}\{B_i = b\}\mathbb{P}\{A_i = a \mid B_i = b\}$, with

$$\mathbb{P}\{B_i = 1\}, \mathbb{P}\{A_i = 1 \mid B_i = 1\}, \mathbb{P}\{A_i = 1 \mid B_i = 0\} \stackrel{\text{i.i.d.}}{\sim} \text{Uniform}(0, 1).$$

Write down the joint density on $(\theta_{00}, \theta_{01}, \theta_{10}, \theta_{11})$ under Model 2. (**3 points**)

- c) Under each model calculate the marginal density of $\mathbb{P}\{A_i = 1\}$. For Model 2, is this marginal density uniform on $(0, 1)$? (**3 points**)
- d) Given data with counts $N_{00}, N_{01}, N_{10}, N_{11}$, write down the posterior density of $(\theta_{00}, \theta_{01}, \theta_{10}, \theta_{11})$ under each model. (**4 points**)
- e) Create a model selection mixture

$$I \sim \text{Uniform}\{1, 2\}, \quad (\theta_{00}, \theta_{01}, \theta_{10}, \theta_{11}) \mid I = i \sim \text{Model } i,$$

with the i.i.d. observations conditionally distributed according to $(\theta_{00}, \theta_{01}, \theta_{10}, \theta_{11})$. Calculate the posterior distribution of I given the observations. (**4 points**)

- f) For the following contingency table, calculate the posterior distribution of I and show that the posterior probability of Model 1 is approximately 3.8 times that of Model 2. (**3 points**)

	$B = 0$	$B = 1$	Row Total
$A = 0$	760	190	950
$A = 1$	5	45	50
Column Total	765	235	1000

- g) **Bonus.** This example is adapted from MacKay (Exercise 35.5, p. 447). MacKay presents it as an illustration of attempts to infer the “direction of causation” using observational data. Model 1 is denoted $A \rightarrow B$ because it specifies the marginal distribution of A and conditional $B \mid A$. Model 2 is denoted $B \rightarrow A$. Discuss whether the 3.8-to-1 posterior odds genuinely provide evidence about causal direction. (**3 bonus points**)